



EAST WEST UNIVERSITY

Department of Computer Science & Engineering

Graph-Based sEMG Gesture Recognition Using NinaPro DB2
TASK 3 REPORT: Improvement, Ablation, Cross-Validation and Explainability

Submitted To

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Date of Submission: 28 August, 2026

1. Task Overview

Task 3 is the validation stage of the project. Instead of assuming that a more complex GNN must be better, we tested seven controlled variants on a development split: repetitions 1, 3 and 4 for training and repetition 6 for validation. Binary Edges produced the best validation Macro-F1 (69.17%) at epoch 36 and was 1.52 points above the No Graph control. We then retrained that selected configuration using repetitions 1, 3, 4 and 6 and evaluated it once on the untouched repetitions 2 and 5. The final model reached 80.37% accuracy and 80.20% Macro-F1. This is 2.81 Macro-F1 points above XGBoost, although the Task 2 first GNN still has the highest single held-out Macro-F1 at 80.98%. Across five grouped folds, the final GNN achieved 67.13 +/- 1.65% Macro-F1 compared with 65.10 +/- 2.11% for XGBoost, and the one-sided Wilcoxon test gave $p = 0.03125$. We finally applied SHAP, LIME and electrode attention to one correct and one wrong prediction from the exact saved checkpoint.

2. Controlled Ablation and Model Selection

Variant	Val. Accuracy	Val. Macro-F1	Best epoch	Parameters
Binary Edges	69.48%	69.17%	36	2,677,749
Top-3 Graph	68.06%	67.82%	27	2,677,749
No Graph	67.85%	67.65%	39	2,677,749
Two GAT Blocks	67.36%	67.07%	21	2,311,981
First ST-EGAT	67.05%	66.91%	20	2,677,749
Final ST-EGAT+	67.17%	66.89%	20	2,890,968
GraphSAGE	66.55%	66.31%	25	1,948,704

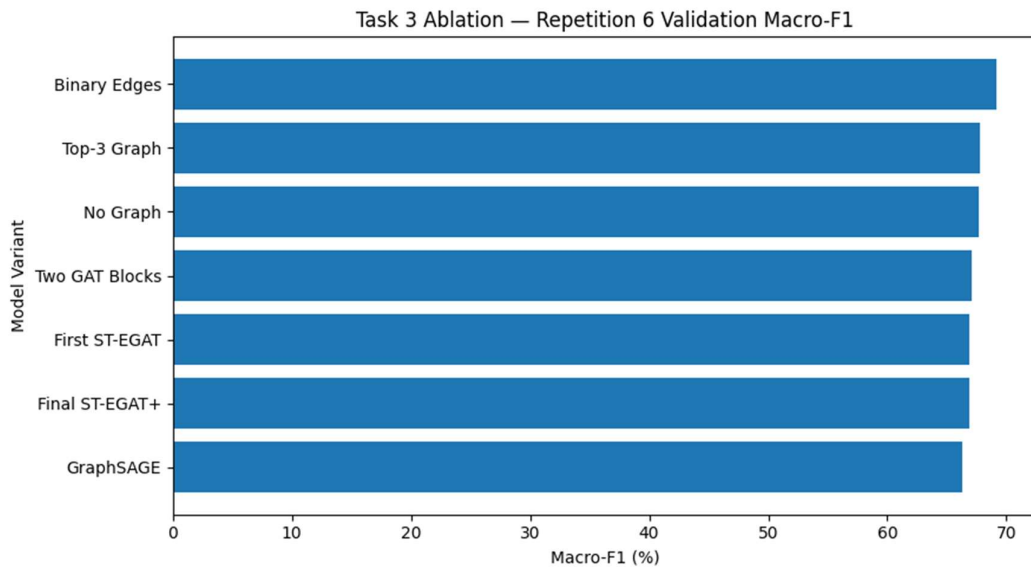


Figure 1. Validation Macro-F1 across the seven Task 3 variants.

Binary Edges gives the best validation Macro-F1. Its 1.52-point advantage over No Graph is direct evidence that message passing contributes useful information in this controlled comparison.

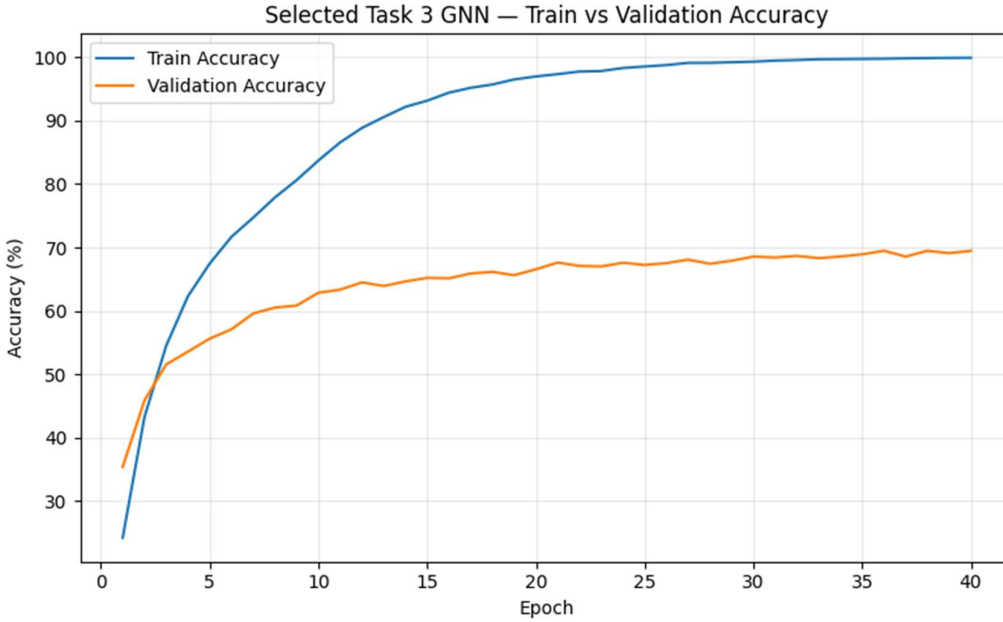


Figure 2. Selected Task 3 model — training and validation accuracy.

The model fits the training repetitions very well, while validation performance levels off earlier. This train-validation gap shows a clear overfitting tendency, which is why the final configuration is chosen by validation performance rather than by training accuracy.

The ablation results are more informative than a simple 'bigger is better' story. Binary connectivity performs better than signed weighted edges on the development split, suggesting that the identity of the strongest electrode connections is more robust here than forcing the network to use the exact signed correlation magnitude. The larger ST-EGAT+ and the GraphSAGE variant also fail to beat the selected model. This is useful evidence that additional complexity does not automatically improve generalization.

3. Final Held-Out Test Performance

Model	Accuracy	Macro-F1	ROC-AUC	PR-AUC	Training time
XGBoost Baseline	77.69%	77.39%	98.25%	85.99%	308.4 s
Task 2 First ST-EGAT	81.16%	80.98%	97.66%	86.93%	3495.6 s
Task 3 Binary Edges	80.37%	80.20%	97.59%	86.38%	2127.9 s

On the held-out test repetitions, the selected Binary-Edge ST-EGAT remains clearly stronger than XGBoost: Macro-F1 rises from 77.39% to 80.20%. At the same time, the Task 2 first GNN retains the highest single held-out Macro-F1 at 80.98%. We therefore do not claim that Task 3 improved the one-shot test score over Task 2. The improvement in Task 3 is methodological: the graph is tested directly, the edge representation is compared, the model is selected without using the final test set, and stability is checked across grouped folds.

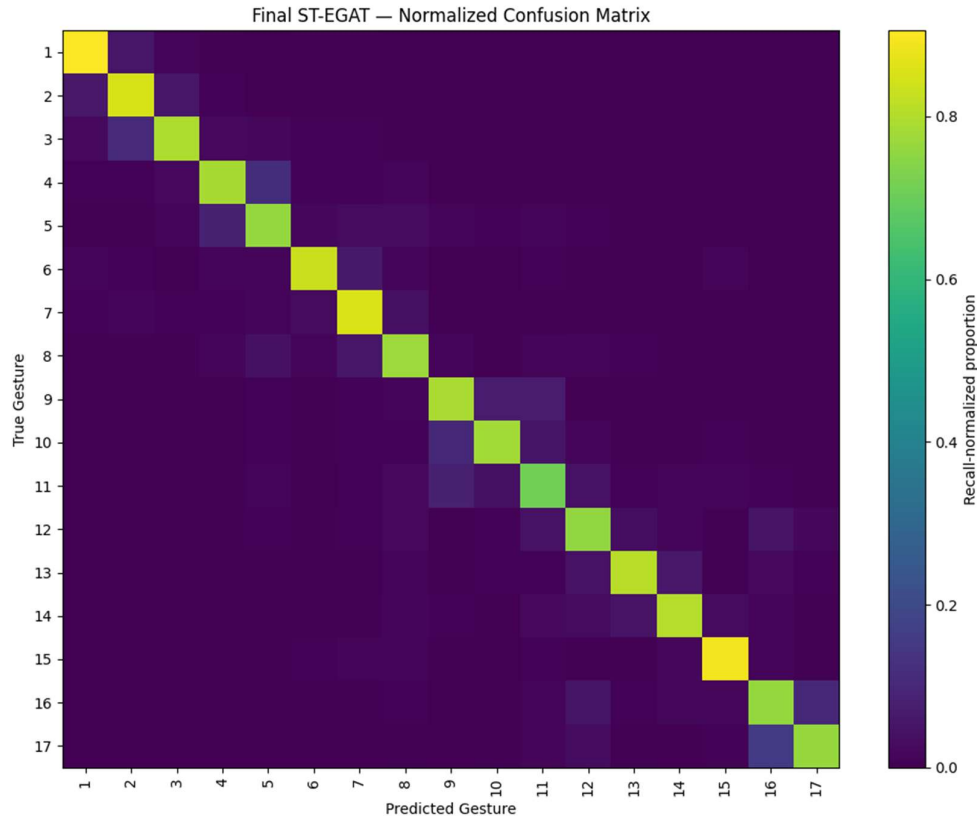


Figure 3. Final Binary-Edge ST-EGAT normalized confusion matrix.

The final confusion matrix remains strongly diagonal across the 17 gestures. The remaining errors are concentrated in a smaller subset of gestures rather than showing a broad failure across classes.

4. Five-Fold Grouped Cross-Validation

Fold	XGBoost Macro-F1	Final GNN Macro-F1
1	67.56%	68.72%
2	63.48%	65.96%
3	62.49%	65.07%
4	65.37%	67.12%
5	66.58%	68.79%

Model	Accuracy	Macro-F1	ROC-AUC	PR-AUC
XGBoost	65.39 $\pm$ 2.08%	65.10 $\pm$ 2.11%	95.44 $\pm$ 0.50%	71.92 $\pm$ 2.18%
Task 3 Binary Edges	67.32 $\pm$ 1.66%	67.13 $\pm$ 1.65%	93.47 $\pm$ 0.67%	70.81 $\pm$ 2.02%

Significance test. Wilcoxon signed-rank statistic = 15.0; one-sided p-value = 0.03125. At $\alpha = 0.05$, the paired fold-wise Macro-F1 evidence supports the final GNN being greater than XGBoost.

The fold-wise result is consistent: the final GNN has higher Macro-F1 than XGBoost in all five paired folds and also shows a lower Macro-F1 standard deviation. XGBoost still has the higher average ROC-AUC and PR-AUC, so the graph model is not uniformly better on every ranking metric. The Wilcoxon result ($p = 0.03125$) supports the Macro-F1 improvement across the paired folds at $\alpha = 0.05$.

5. Explainability of the Final Checkpoint

Before generating explanations, the explainability notebook reconstructs the exact Binary-Edge checkpoint from epoch 36 and reproduces the 80.37% accuracy and 80.20% Macro-F1 result. SHAP and LIME are applied to the top selected global features for one correct and one wrong sample, while electrode attention and active graph connections provide a complementary view of what the graph branch is using. These outputs are treated as model-attribution evidence, not as causal physiological proof.

Sample	Source	Actual → Pred.	Confidence	Top SHAP	Top electrode	Top LIME condition
Correct	S31 / rep 5	G13 → G13	99.92%	ch11_WAMP	Ch12	ch11_WAMP $\leq$ -0.36
Wrong	S26 / rep 5	G13 → G14	99.69%	ch11_WAMP	Ch1	ch11_RMS $>$ -0.07

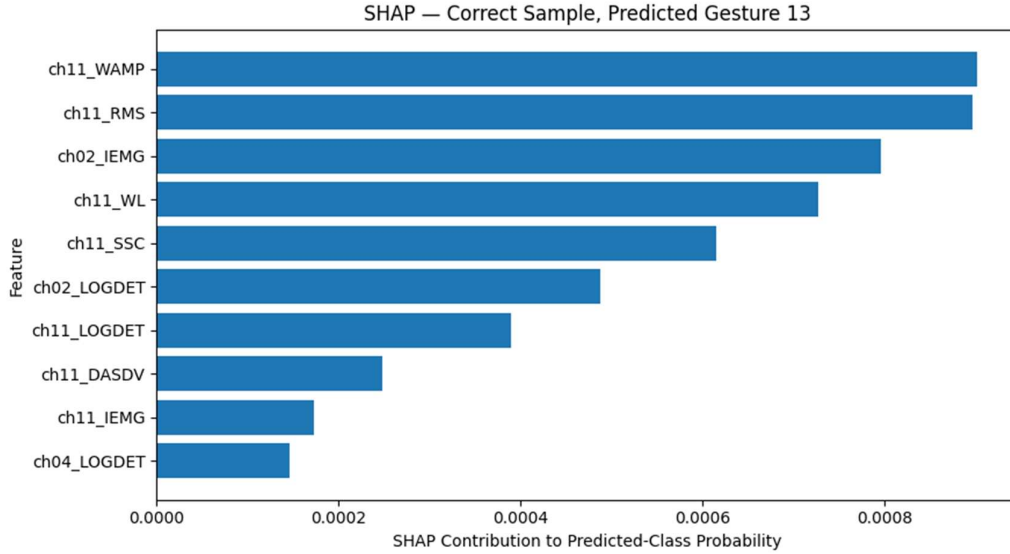


Figure 4. SHAP attribution for the selected correct sample (predicted Gesture 13).

For the selected correct prediction, *ch11_WAMP* is the strongest SHAP feature and Channel 12 receives the highest graph-readout attention.

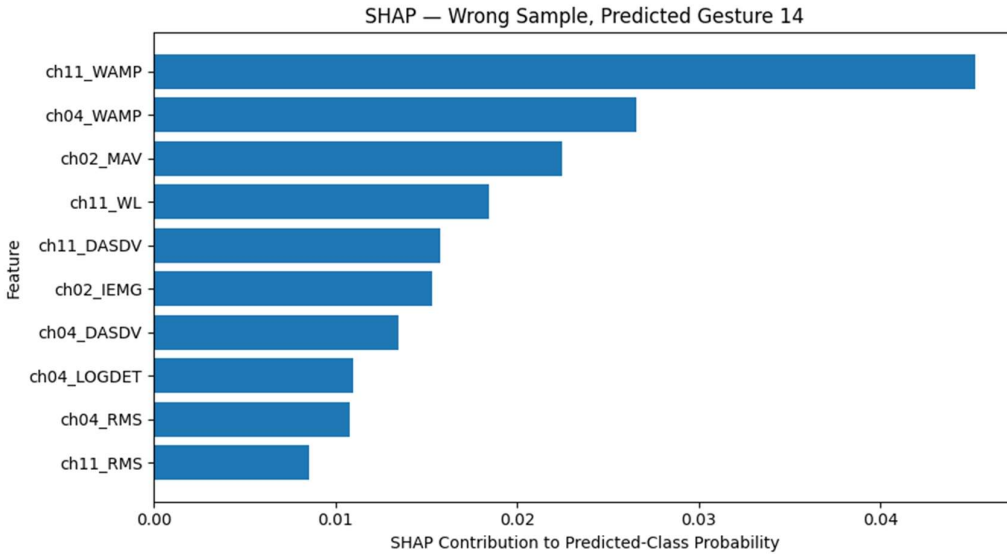


Figure 5. SHAP attribution for the selected wrong sample (actual Gesture 13, predicted Gesture 14).

The wrong prediction is also influenced strongly by Channel-11 descriptors, but its electrode attention shifts toward Channel 1. Because the mistake is made with high confidence, it is a useful example of where the model can still be misleading.

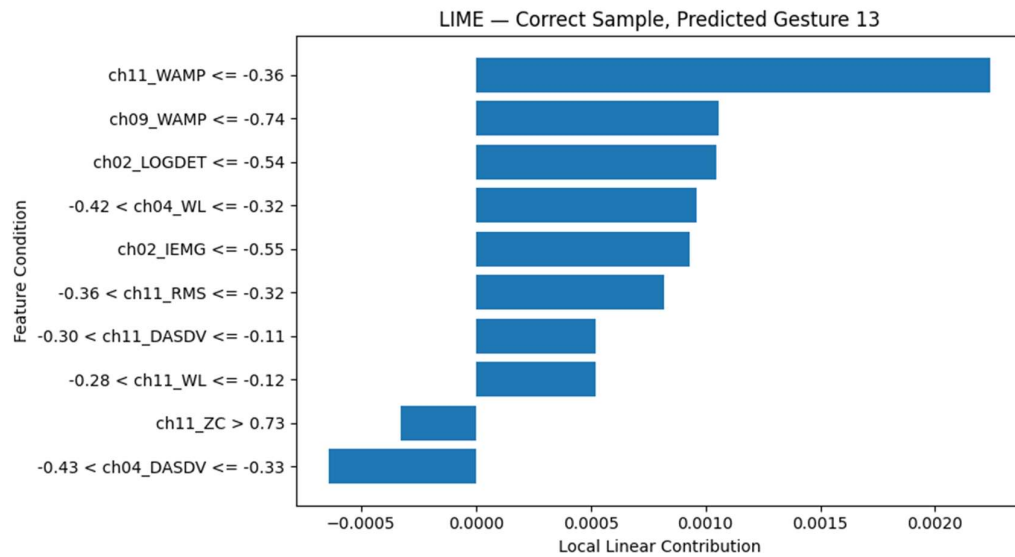


Figure 6. LIME local contributions for the correct sample.

For the correct sample, LIME identifies a low ch11_WAMP condition as the strongest local positive contribution toward Gesture 13.

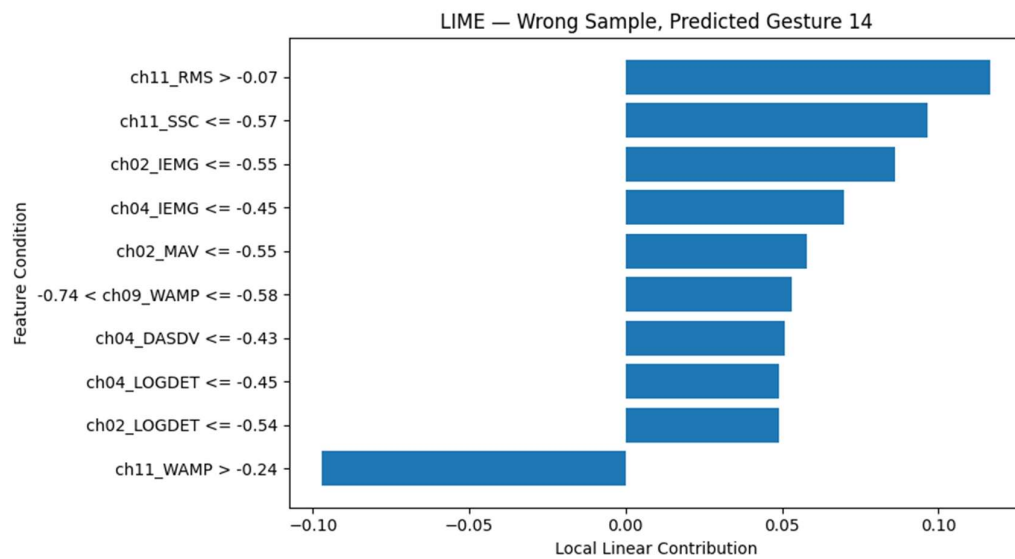


Figure 7. LIME local contributions for the wrong sample.

For the wrong sample, ch11_RMS and ch11_WAMP are among the largest local contributors, again showing the importance of Channel-11 activity in the model's decision.

6. Fair Context Against Related Work

Study	Protocol context	Representative result	Comparability
Current Task 3	NinaPro DB2, Exercise B, 17 gestures, repetition split	80.37% Accuracy / 80.20% Macro-F1	Direct result for this project.
MSFF-Net (Peng et al., 2022)	NinaPro DB2, 49 active gestures, repetitions 1/3/4/6 vs 2/5	87.02% Accuracy	Same repetition pattern but different gesture scope and non-GNN architecture; useful context, not a like-for-like score.
Xu et al. (2024)	HD-sEMG, 11 subjects, 17 gestures	68.0% user-independent; 92.3% with transfer learning	Different sensor density and cross-user protocol.
Lee et al. (2025)	NinaPro DB5 and other EMG datasets	92.4% inter-subject on DB5 for one graph setting	Different NinaPro database, sensor count and task.

Published results are included as context, not as direct state-of-the-art claims. The closest exact-dataset benchmark uses the same repetition pattern but a different gesture scope and a non-GNN architecture, while other graph studies use different NinaPro databases, sensor densities or cross-user protocols. For that reason, the fairest performance comparison in this project is against XGBoost and the Task 2 GNN, which share the exact same preprocessing, windowing, gesture subset and repetition split.

7. Conclusion

Task 3 gives the clearest evidence in the project about what the graph is contributing. The selected Binary-Edge ST-EGAT is chosen by validation, not by test-set cherry-picking; it beats the no-graph control on the development split and outperforms XGBoost in Macro-F1 across all five grouped folds. The held-out score is slightly below the Task 2 first GNN, so the final claim is deliberately modest: the graph contribution is supported by ablation and cross-validation, but more complexity did not produce a higher single test score. SHAP, LIME and electrode attention add a final layer of evidence by showing which engineered features and electrodes influenced both a correct and a confidently wrong prediction. Taken together, the results provide a balanced and defensible final evaluation rather than a score-only comparison.